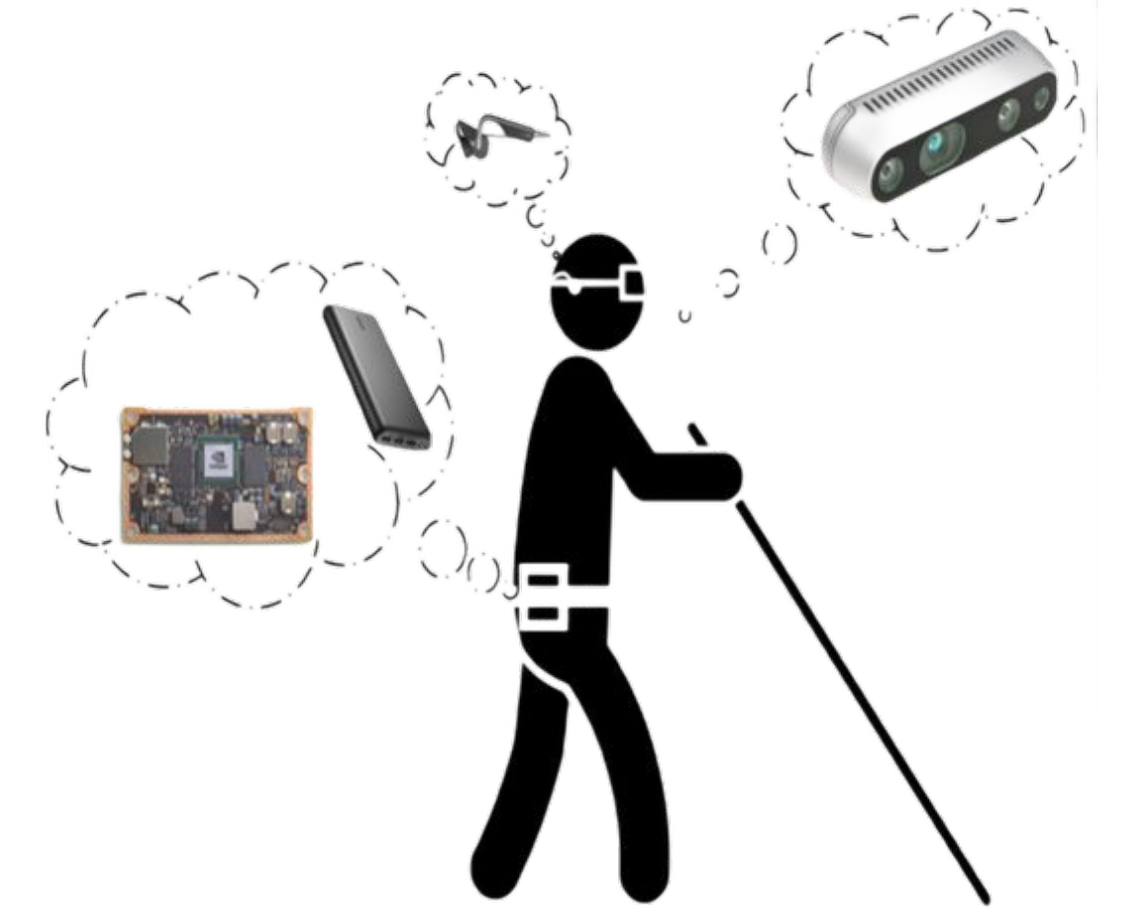
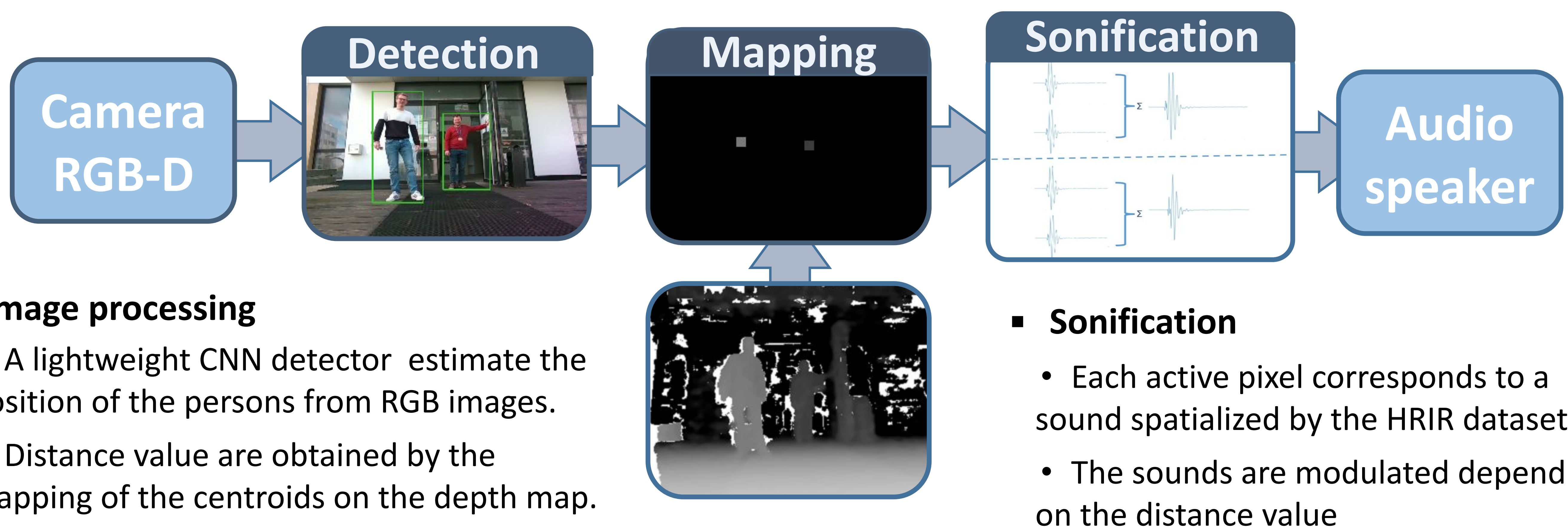


Introduction

- **Context:** *Mobility* and *social Interaction* tasks in an unknown environment are difficult tasks for visual impaired (V.I). Assistive method are developed to improve scene understanding by providing relevant information to enable safe navigation.
- **Objectif:** A real-time visuo-auditory substitution assistive device focuses on 3-dimension person localization in order to ease urban walking.



Method



Real-time Implementation

- **Laptop**
 - A YOLOv5-small CNN detector estimate the position of the persons from RGB images.
 - System performance evaluation on the CPU or GPU

	CPU	GPU	
		LibTorch	TensorRT
Yolov5-small (ms)	135	16.3	7.5
Global processing (ms)	142	23.6	15.3

Processing time of the YOLOv5-small using a laptop with the overall system

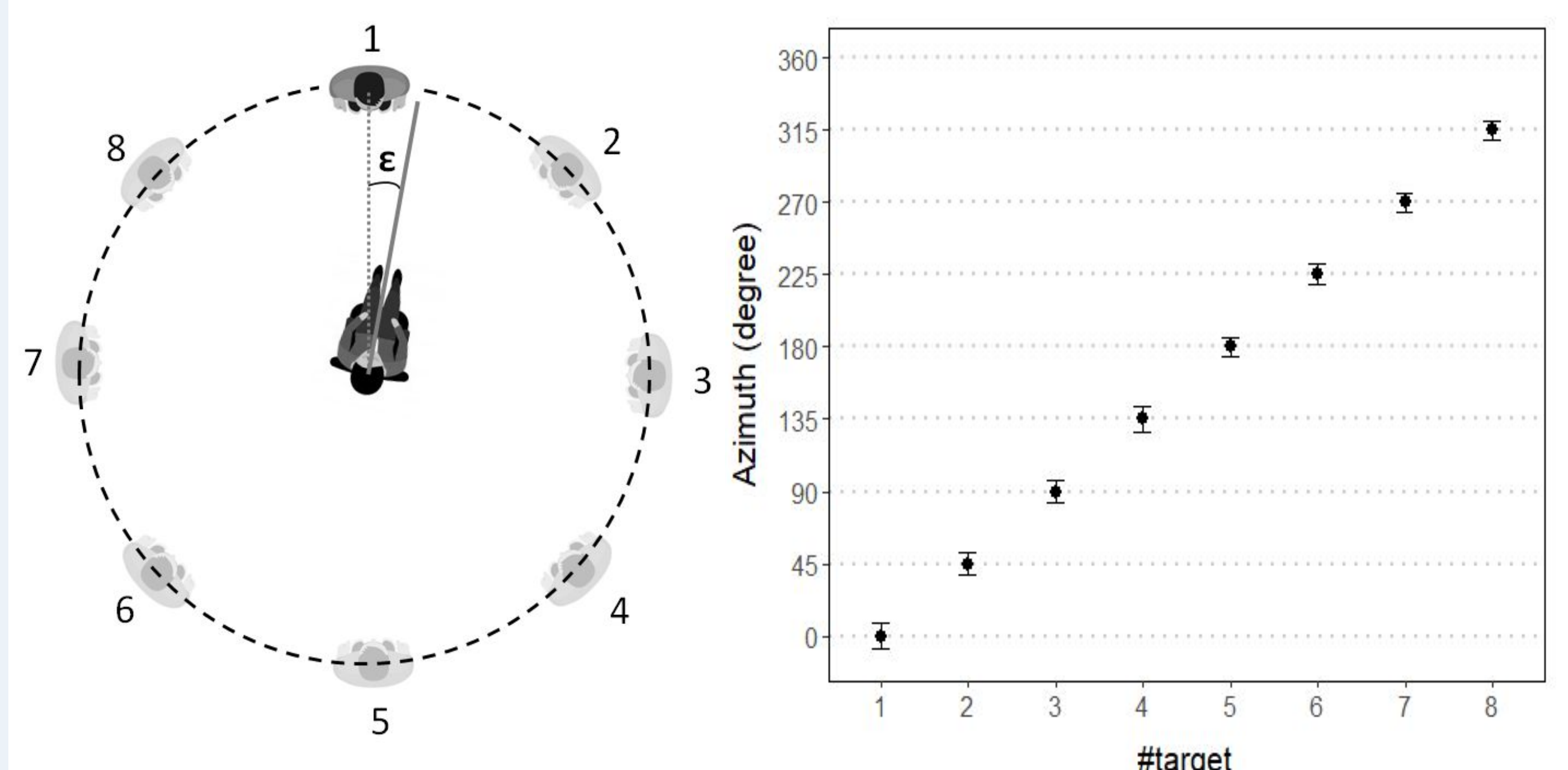
- **Embedded board**
 - Implemented on a Nvidia Jetson TX2 with two power modes and two dynamics ranges

	FP32 15W	FP32 7.5W	FP16 15W	FP16 7.5W
Yolov5-small (ms)	55	71	35	47
Global Processing (ms)	62	80	43	56

Processing time of the neural network on the embedded target in comparison with the overall system

Experimentation

- A blindfolded participant, placed in the center of the circle, tries to locate a standing person.
- Evaluation of the azimuth angular error between the user's prediction and the target's position.



Experimental setup

Azimuth of each target with the associated mean value of the azimuth angular error

Conclusion

- A lightweight and robust A.I. method to localize a person or an object by sonification.
- A function wearable and low-energy consumption prototype for detection of person which operate close to real-time.
- A 3-Dimension localization of person by spatialized and modulated stereophonic sounds
- Human capacity to perceive a person's position has been evaluated and demonstrated experimentally.